

iStar-DT-X ver 1.0

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iStarDT-X Grammar (1/3)

Primitives, Enums

```
[symbol]      ::= Any valid XML character except quotes and angle brackets
[letter]      ::= "A" | "B" | ... | "Z" | "a" | "b" | ... | "z"
[letter-small] ::= "a" | "b" | ... | "z"
[digit]       ::= "0" | "1" | ... | "9"
[character]   ::= [letter] | [digit] | [symbol]
[alphanum]    ::= [letter] | [digit] | "."
[string]      ::= [character]+
[identifier]  ::= [letter-small][alphanum]*
[boolean]     ::= "true" | "false"
[decimal]     ::= [digit]+ [ "." [digit]+ ]
[integer]     ::= [digit]+
[del-style]   ::= 'sync' | 'async'
```

Boolean Expressions

```
[boolean-expression] ::= [bool-const] | [atom] | [previous-bool] |
[bool-const]         ::= "<boolConst>" [boolean] "</boolConst>"
[atom]               ::= [predicate-ref] | {condition-id}+
                        [goal-ref] | [task-ref] | {effect-id}
{predicate-id}       ::= "<predicateID>" [identifier] "</predicateID>"
{condition-id}       ::= "<conditionID>" [identifier] "</conditionID>"
{effect-id}          ::= "<effectID>" [identifier] "</effectID>"
{goal-id}             ::= "<goalID>" [identifier] "</goalID>"
{task-id}             ::= "<taskID>" [identifier] "</taskID>"
[previous-bool]      ::= "<previous>" [atom] "</previous>"
[boolean-operation]  ::= [and] | [or] | [not]
[and]                ::= "<and>" [boolean-expression] [boolean-expression]+ "</and>"
[or]                 ::= "<or>" [boolean-expression] [boolean-expression]+ "</or>"
[not]                ::= "<not>" [boolean-expression] "</not>"
[comparison-op]      ::= "gt" | "gte" | "lt" | "lte" | "eq" | "neq"
[numeric-comparison] ::= "<" [comparison-op] ">"
                    "<left>" [numeric-expression] "</left>"
                    "<right>" [numeric-expression] "</right>"
                    "</>" [comparison-op] ">"
```

Numeric Expressions

```
[numeric-expression] ::= [num-const] | [num-atom] | [num-previous] |
                        [arithmetic-operation]
[num-const]           ::= "<numConst>" [decimal] "</numConst>"
[num-atom]            ::= [atom] | [variable-id] | [qual-id]
[variable-id]         ::= "<variableID>" [identifier] "</variableID>"
[qual-id]             ::= "<qualID>" [identifier] "</qualID>"
[num-previous]        ::= "<previous>" [num-atom] "</previous>"
[arithmetic-operator] ::= [add] | [subtract] | [multiply] | [divide]
[add]                ::= "<add>" [numeric-expression] [numeric-expression]+ "</add>"
[multiply]            ::= "<multiply>" [numeric-expression] [numeric-expression]+
                        "</multiply>"
[subtract]            ::= "<subtract>" "<left>" [numeric-expression] "</left>"
                        "<right>" [numeric-expression] "</right>" "</subtract>"
[divide]              ::= "<divide>" "<left>" [numeric-expression] "</left>"
                        "<right>" [numeric-expression] "</right>" "</divide>"
```

Parameters

```
[parameters] ::= "<parameters>" [parameter] "</parameters>"
[parameter]  ::= "<parameter>" [param-type] [param-name] "</parameter>"
[param-type] ::= "<param-type>" [identifier] "</param-type>"
[param-name] ::= "<param-name>" [identifier] "</param-name>"
```

Figure 1: BNF specification of iStarDT-X: primitives and expressions.

iStarDT-X Grammar (2/3)

Root

```

[istardt-model] ::= "<iStarDT " [namespaces] ">" [model-header] [options] [actors]
                  "</iStarDT>"
[namespaces]    ::= " xmlns=" [string] "' " [" xmlns:xsi=" [string] "'"]
                  [" xsi:schemaLocation=" [string] "'"]
[model-header]  ::= "<header title = " [string] "' author = " [string]
                  "' source = " [string] "' lastUpdated=" [string] "'>"
                  [string] "</header>"
[options]       ::= "<options continuous = " [boolean] "'
                  infeasibleActionPenalty = " [decimal] "'>"
                  "</options>"
[actors]        ::= "<actors>" [actor]+ "</actors>"
[actor]         ::= "<actor " [actor-attributes] ">" [actor-content] "</actor>"
[actor-attributes] ::= "name=" [identifier] "' " [description=" [string] "' ]
[actor-content]  ::= [ [goals] ] [ [tasks] ] [ [qualities] ] [ [predicates] ]
                  [ [variables] ] [ [condBoxes] ] [ [cross-runs] ]
                  [ [exported-set] ] [ [initializations] ] [ [definitions] ]

```

Goals

```

[goals]         ::= "<goals>" [goal]+ "</goals>"
[goal]          ::= "<goal name=" [identifier] "' " [" root=" [boolean] "'"]
                  [" description=" [string] "'"] [" terminal=" [boolean] "'"]
                  [" episodeLength=" [integer] "'"] [" actor=" [identifier] "'"]
                  [" delegation-style=" [del-style] " ] ">" [goal-content] "</goal>"
[goal-content]  ::= [ [parameters] ] [ [pre] ] [ [npr] ] [ [refinement] ]
[pre]          ::= "<pre>" [boolean-expression] "</pre>"
[npr]          ::= "<npr>" [boolean-expression] "</npr>"
[refinement]    ::= "<refinement type=" [ref-type] "'>"
                  [ [ref-condition] ]
                  ([childGoal] | [childTask])+ "</refinement>"
[ref-condition] ::= "<ref-condition>" [pick-cond] | [forall-cond] "</ref-condition>"
[pick-cond]    ::= "<pick>" [cond-content] "</pick>"
[forall-cond]  ::= "<forall>" [cond-content] "</forall>"
[cond-content] ::= "<params>" [cond-param]+ "</params>"
                  "<condition>" [cond-pred] "</condition>"
[cond-param]   ::= "<param>" [identifier] "</param>"
[cond-pred]    ::= "<content>" [identifier] "</content>"
[ref-type]     ::= "AND" | "OR"
[childGoal]    ::= "<childGoal ref=" [identifier] "'/" /* reference to goal name */
[childTask]    ::= "<childTask ref=" [identifier] "'/" /* reference to task name */
[goal-ref]     ::= "<goal name=" [identifier] "'>" [parameters] "</goal>"

```

Tasks

```

[tasks]         ::= "<tasks>" [task]+ "</tasks>"
[task]          ::= "<task name=" [identifier] "' " [" description=" [string] "'"]
                  [" actor=" [identifier] "'"] [" delegation-style=" [del-style] " ]
                  ">" [task-content] "</task>"
[task-content]  ::= [ [parameters] ] [effectGroup] [ [pre] ] [ [npr] ]
[effectGroup]   ::= "<effectGroup>" [effect]+ "</effectGroup>"
[effect]        ::= "<effect name=" [identifier] "' " [ " satisfying=" [boolean] "'"]
                  [" probability=" [decimal] "'"] [ " description=" [string] "'"]
                  ">" [effect-content] "</effect>"
[effect-content] ::= [ [turnsTrue]* ] [ [turnsFalse]* ] [ [sets]* ] [ [pre] ] [ [npr] ]
[turnsTrue]     ::= "<turnsTrue>" [identifier] "</turnsTrue>"
[turnsFalse]    ::= "<turnsFalse>" [identifier] "</turnsFalse>"
[sets]          ::= "<set>" [variable-id] [num-const] "</set>"
[task-ref]      ::= "<task name=" [identifier] "'>" [parameters] "</task>"

```

Figure 2: BNF specification of iStarDT-X: Models, Goals, and Tasks.

iStarDT-X Grammar (3/3)

Qualities

```
[qualities] ::= "<qualities>" [quality]+ "</qualities>"
[quality]  ::= "<quality name='\" [identifier] '\" description='\" [string] '\"\"\"
               [\" root='\" [boolean] '\"\"\" ]>"
               [numeric-expression] "</quality>"
```

Predicates and Variables

```
[predicates] ::= "<predicates>" [predicate]+ "</predicates>"
[predicate]  ::= "<predicate name = '\" [identifier] '\" description = '\" [string] '\">"
               [ [parameters] ] "</predicate>" % see above
[predicate-ref] ::= "<predicate name = '\" [identifier] '\">"
                  [ [parameters] ] "</predicate>" % see above
[variables]  ::= "<variables>" [variable]+ "</variables>"
[variable]  ::= "<variable description = '\" [string] '\">"
               [identifier] "</variable>"
```

Condition Boxes

```
[condBoxes] ::= "<condBoxes>" [condBox]+ "</condBoxes>"
[condBox]  ::= "<condBox name='\" [identifier] '\" description='\" <string> '\"\"\" ]>"
               [predicate-ref] "</condBox>"
```

Cross-runs, Exported Sets, Initializations, Defininos

```
[cross-runs] ::= "<crossRuns>" [cross-run]+ "</crossRuns>"
[cross-run]  ::= "<crossRun>" [predicate-id] | [variable-id] | [qual-id]
               "</crossRun>"
[exported-set] ::= "<exportedSet>" [export]+ "</exportedSet>"
[export]      ::= "<export continuous=" [boolean] [ "minVal=" [decimal]
               [ "maxVal=" [decimal] ] ">" [exp-id] "</export>"
[exp-id]      ::= [goal-id] | [task-id] | [predicate-id] | [variable-id] | [qual-id]
[initializations] ::= "<initializations>" [initialization] "</initialization>"
[initialization] ::= [b-init] | [d-init]
[b-init]      ::= "<initialization element=" [predicate-id] ">"
               [boolean] "</initialization>"
[d-init]      ::= "<initialization element=" [cont-init-id] ">"
               [decimal] "</initialization>"
[cont-init-id] ::= [variable-id] | [qual-id]
[definitions] ::= "<definitions>" [definition] "</definitions>"
[definition]  ::= "<definition>" [definiendum] [definiens] "</definition>"
[definiendum] ::= "<definiendum name = '\" [identifier] '\">"
               [ [parameters] ] "</definiendum>"
[defiens]     ::= "<definiens>" [boolean-expression] "</definiens>"
```

Figure 3: BNF specification of iStarDT-X: Qualities, Predicates, Variables, Condition Boxes, Cross-run, Exported and Initialization sets.